ICT 6101: Multiple Choice Question Bank

150 MCQs covering Classes 1 to 5 (Introduction, Vulnerability, Threat and Malware, Cloud Computing, Cloud Security)

How to use: each set has 10 questions with four options (a to d). Choose one answer, then check the answer key that follows each set. The explanation in the key gives the reason, based on the class slides. Coverage: Class 1 (CIA triad, assets, threats, CMM dimensions, ICT indices); Class 2 (vulnerability, CVE, CVSS, threats, attack vectors and surface, risk, incidents); Class 3 (vulnerability causes and scanning, patching, insider threats, viruses, worms, trojans, rootkits, bots, malware economics); Class 4 (cloud definition, characteristics, service and deployment models, technologies, drivers, edge and fog); Class 5 (cloud challenges, eleven threats, component attacks, incidents).

SET 1 (Questions 1 to 10)

1. Which of the following is NOT a principle of the CIA triad?

a) Authentication
b) Confidentiality
c) Integrity
d) Availability

2. Which of these is listed in the slides as a confidentiality measure?

a) Checksums
b) Security tokens
c) Uninterrupted power supplies
d) Data backups

3. Which of these is a standard measure to guarantee availability?

a) Checksums
b) Two-factor authentication
c) Data redundancy
d) Biometric verification

4. Which of these is an information asset?

a) A phishing email
b) A fire in the server room
c) An unlocked office door
d) A database of customer records

5. Which of these is an intentional threat?

a) Criminal hacking
b) Hard drive failure
c) A natural flood
d) Forgetting to update antivirus

6. Which threat source is motivated mainly by economic gain and uses phishing, spam, and spyware?

a) Nation states
b) Criminal groups
c) Terrorist organizations
d) Trusted vendors

7. A malicious insider is best described as:

a) An outsider with no system access
b) A government intelligence officer
c) An employee or partner with legitimate access who abuses it
d) A software bug in a database

8. Cyberspace is best described as:

a) A single computer network inside one building
b) A type of malware
c) A physical server room
d) An intricate environment of interactions among people, software, and services

9. How many dimensions does the Cybersecurity Capacity Maturity Model (CMM) for Nations define?

a) 5
b) 3
c) 4
d) 6

10. Which dimension covers legal and regulatory frameworks?

a) Dimension 1
b) Dimension 4
c) Dimension 2
d) Dimension 5

Answer Key: Set 1 (Questions 1 to 10)

Q	Answer	Why
1	**a) Authentication**	The CIA triad is Confidentiality, Integrity, Availability.
2	**b) Security tokens**	Confidentiality measures: encryption, 2FA, biometrics, security tokens.
3	**c) Data redundancy**	Availability measures: backups, firewalls, backup power, data redundancy.
4	**d) A database of customer records**	Information assets are databases and physical files holding sensitive data.
5	**a) Criminal hacking**	Intentional threats include criminal hacking and malicious insiders.
6	**b) Criminal groups**	Criminal groups seek economic benefit through phishing, spam, spyware, and malware.
7	**c) An employee or partner with legitimate access who abuses it**	Malicious insiders have legitimate access and abuse their privileges.
8	**d) An intricate environment of interactions among people, software, and services**	Cyberspace is the environment formed by ICT devices, networks, and their users.
9	**a) 5**	The CMM defines five dimensions.
10	**b) Dimension 4**	Dimension 4 is Legal and Regulatory Frameworks.

SET 2 (Questions 11 to 20)

11. The term 'cyber' is best described as:

a) A type of firewall
b) A certified security standard
c) A combining form meaning computer or computer network
d) A form of encryption

12. The National Cyber Security Index (NCSI) was introduced by:

a) International Telecommunication Union
b) World Economic Forum
c) United Nations
d) e-Governance Academy Foundation, Estonia

13. Which index is introduced by the World Economic Forum?

a) Networked Readiness Index (NRI)
b) ICT Development Index (IDI)
c) E-Government Development Index (EGDI)
d) Global Cybersecurity Index (GCI)

14. Which index is introduced by the United Nations?

a) Networked Readiness Index (NRI)
b) E-Government Development Index (EGDI)
c) Global Cybersecurity Index (GCI)
d) National Cyber Security Index (NCSI)

15. How many indicators does the ICT Development Index (IDI) depend on?

a) 5
b) 8
c) 11
d) 15

16. According to the slides, Bangladesh ranked which position in the GCI 2024?

a) 89th out of 133 countries
b) 100th out of 193 countries
c) 48th in the NCSI
d) 53rd out of 194 countries

17. Which of these is NOT a component of the NRI?

a) Cyber insurance
b) Environment for ICT
c) Stakeholders (individuals, businesses, government)
d) Usage of ICT

18. How many pillars does the Global Cybersecurity Index (GCI) have?

a) 3
b) 5
c) 4
d) 11

19. A botnet launching a DDoS attack on a website is best described as a:

a) Cyber threat
b) Cyber risk
c) Cyber attack
d) Vulnerability

20. 'The possibility that a particular attack may occur' is the definition of:

a) Cyber attack
b) Cyber risk
c) Asset
d) Cyber threat

Answer Key: Set 2 (Questions 11 to 20)

Q	Answer	Why
11	**c) A combining form meaning computer or computer network**	Cyber means computer, computer network, or virtual reality.
12	**d) e-Governance Academy Foundation, Estonia**	The NCSI project team sits at the e-Governance Academy Foundation, Estonia.
13	**a) Networked Readiness Index (NRI)**	The WEF introduced the NRI.
14	**b) E-Government Development Index (EGDI)**	The UN introduced the EGDI.
15	**c) 11**	The IDI uses eleven indicators in three categories.
16	**d) 53rd out of 194 countries**	GCI 2024: Bangladesh 53rd of 194.
17	**a) Cyber insurance**	NRI components: environment for ICT, stakeholders, and usage.
18	**b) 5**	Legal, technical, organizational, capacity development, and cooperation.
19	**c) Cyber attack**	The attack is the offensive action itself.
20	**d) Cyber threat**	A threat is the possibility of an attack; risk is the estimated loss.

SET 3 (Questions 21 to 30)

21. Which of these is an example of a cyber asset?

a) A code repository
b) A weak password
c) A phishing email
d) A misconfigured firewall

22. CISA defines a vulnerability as:

a) A person who attacks a system
b) A set of conditions that may lead to failure of confidentiality, integrity, or availability
c) The likelihood of a financial loss
d) A list of known malware

23. Which of these is a human vulnerability?

a) A buffer overflow bug
b) A misconfigured firewall
c) Falling for smishing
d) An unpatched kernel

24. According to the 2022 Positive Technologies data in the slides, what share of vulnerabilities related to web application code flaws?

a) 58%
b) 85%
c) 45%
d) 72%

25. CVE stands for:

a) Common Vulnerabilities and Exposures
b) Common Vulnerability Scoring Engine
c) Central Vulnerability Enumeration Database
d) Cyber Vulnerability Evaluation

26. The CVSS score ranges from:

a) 0 to 5
b) 0 to 10
c) 1 to 100
d) A to F

27. The Heartbleed vulnerability (CVE-2014-0160) has a CVSS score of:

a) 9.8
b) 4.3
c) 7.5
d) 10.0

28. In CVSS v3.x, a score from 9.0 to 10.0 is rated:

a) High
b) Medium
c) Low
d) Critical

29. Which CVSS metric group describes how exploit code maturity changes over time?

a) Temporal
b) Base
c) Scope
d) Environmental

30. An exposure, as defined in the slides, is:

a) A flaw attackers use to run code
b) A mistake in code or configuration that gives an attacker access
c) A type of malware
d) A risk score

Answer Key: Set 3 (Questions 21 to 30)

Q	Answer	Why
21	**a) A code repository**	Cyber assets include software such as code repositories.
22	**b) A set of conditions that may lead to failure of confidentiality, integrity, or availability**	This is the CISA definition quoted in Class 2.
23	**c) Falling for smishing**	Human vulnerabilities include phishing, smishing, and quishing.
24	**d) 72%**	72% of vulnerabilities were web application code flaws (2022).
25	**a) Common Vulnerabilities and Exposures**	CVE is the catalog of publicly known vulnerabilities and exposures.
26	**b) 0 to 10**	CVSS scores run from 0.0 to 10.0.
27	**c) 7.5**	The slides give 7.5, which indicates high severity.
28	**d) Critical**	v3.x: High 7.0-8.9, Critical 9.0-10.0.
29	**a) Temporal**	Temporal metrics include exploit code maturity, remediation level, and report confidence.

Q	Answer	Why
30	**b) A mistake in code or configuration that gives an attacker access**	Exposures are mistakes that give access; vulnerabilities are exploitable flaws.

SET 4 (Questions 31 to 40)

31. The risk formula given in the slides is:

a) Risk = Asset + Threat
b) Risk = Vulnerability / Impact
c) Risk = Threat x Vulnerability x Impact
d) Risk = Threat - Vulnerability

32. CVSS should be understood as:

a) A risk score that includes likelihood
b) A threat model
c) A type of vulnerability scanner
d) A severity measure, not a risk score

33. Which risk management option transfers the risk?

a) Buying cyber insurance
b) Accepting the residual risk
c) Applying a security patch
d) Ignoring a low-value risk

34. Which is an external cyber risk?

a) An employee sharing a password
b) A DDoS attack from outside the organization
c) A contractor misusing access
d) A negligent staff member

35. In the Venn model of asset, threat, and vulnerability, the overlapping area represents:

a) The asset
b) The threat
c) Risk
d) The vulnerability

36. The 'universal truth' stated in the risk slides is that:

a) All threats can be removed with antivirus
b) Encrypted data carries zero risk
c) Vulnerabilities only exist in software
d) You cannot eliminate or entirely protect against all threats

37. In the house example, the broken door lock is:

a) A vulnerability
b) A threat
c) An asset
d) A risk

38. In the same house example, the burglar who knows about the broken lock is:

a) A vulnerability
b) A threat
c) An asset
d) A risk

39. The Tesla data leak described in the slides was:

a) A zero-day exploit
b) A DDoS attack
c) An inside job by former employees
d) A nation-state malware attack

40. The MOVEit ransomware incident was caused by:

a) A worm in the email system
b) A faulty CrowdStrike update
c) A rootkit installed by a Trojan
d) Exploiting a known software vulnerability

Answer Key: Set 4 (Questions 31 to 40)

Q	Answer	Why
31	**c) Risk = Threat x Vulnerability x Impact**	Risk combines threat, vulnerability, and impact.
32	**d) A severity measure, not a risk score**	The slides state CVSS is not a risk score or a threat model.
33	**a) Buying cyber insurance**	Transfer moves the financial impact to another party, such as an insurer.
34	**b) A DDoS attack from outside the organization**	External risks come from outside, such as DDoS, phishing, and ransomware.
35	**c) Risk**	Risk exists where a threat can exploit a vulnerability in an asset.
36	**d) You cannot eliminate or entirely protect against all threats**	No system is completely protected against every threat.
37	**a) A vulnerability**	Vulnerabilities are weaknesses; the burglar is the threat.
38	**b) A threat**	A threat is a potential danger that could exploit a vulnerability.
39	**c) An inside job by former employees**	The slides say it was an inside job, with no fancy hacks.
40	**d) Exploiting a known software vulnerability**	The Clop gang exploited a known MOVEit Transfer vulnerability.

SET 5 (Questions 41 to 50)

41. An attack vector is:

a) A path or method used to gain unauthorized access
b) The set of all exploitable points
c) A type of asset
d) A security certification

42. An attack surface is best described as:

a) A single open port
b) The sum of all exploitable points in a system or network
c) The attacker's motivation
d) The list of published CVEs

43. Which of these is a social engineering attack vector?

a) Cryptojacking
b) HTTPS spoofing
c) Tailgating
d) SYN flood

44. Pretexting, in social engineering, means:

a) Following someone into a restricted area
b) Hiding malware in a JavaScript page
c) Flooding a network with traffic
d) Fabricated scenarios used to steal information

45. Which injection attack targets databases through input fields?

a) SQL injection
b) DNS spoofing
c) NTP amplification
d) Wi-Fi eavesdropping

46. NTP amplification belongs to which category of attack vectors?

a) Man-in-the-middle
b) Denial of service
c) Injection
d) Social engineering

47. A key characteristic of a Trojan horse is that it:

a) Replicates across networks without user action
b) Attaches itself to host files
c) Is disguised as a legitimate program
d) Hides only from the operating system

48. A rootkit is best described as malware that:

a) Displays unwanted advertisements
b) Gathers private information for a third party
c) Consumes resources without damage
d) Gives complete control of a system while staying hidden

49. Which of these is NOT a type of malware?

a) Firewall
b) Worm
c) Trojan
d) Adware

50. An attack that lures victims with text messages is called:

a) Vishing
b) Smishing
c) Quishing
d) Pretexting

Answer Key: Set 5 (Questions 41 to 50)

Q	Answer	Why
41	**a) A path or method used to gain unauthorized access**	Attack vectors are paths; the attack surface is the total set of exploitable points.
42	**b) The sum of all exploitable points in a system or network**	The attack surface aggregates all points an attacker could target.
43	**c) Tailgating**	Tailgating means following someone into a restricted area.
44	**d) Fabricated scenarios used to steal information**	Pretexting invents a believable story to extract information.
45	**a) SQL injection**	SQL injection submits malicious SQL through application input.
46	**b) Denial of service**	It is a DoS technique that amplifies traffic.

Q	Answer	Why
47	**c) Is disguised as a legitimate program**	Trojans appear legitimate but hide malicious functions.
48	**d) Gives complete control of a system while staying hidden**	Rootkits provide hidden, root-level control.
49	**a) Firewall**	A firewall is a defensive security tool, not malware.
50	**b) Smishing**	Smishing is phishing by SMS; vishing is voice phishing; quishing uses QR codes.

SET 6 (Questions 51 to 60)

51. Which of these is NOT listed as a cause of vulnerabilities?

a) Software bugs
b) Unchecked user input
c) Strong, unique passwords for every account
d) Familiarity with common software

52. Non-credentialed scans are limited because they:

a) Log in with administrator credentials
b) Always find every internal flaw
c) Require software on each device
d) Miss most vulnerabilities inside the target environment

53. Penetration testing is best described as:

a) An exercise that tries to exploit vulnerabilities to measure real impact
b) A passive list of known CVEs
c) A software patch installation process
d) A type of data backup

54. Which statement correctly distinguishes vulnerability assessment from penetration testing?

a) Vulnerability assessment exploits vulnerabilities
b) Penetration testing exploits vulnerabilities
c) Penetration testing is always fully automated
d) Vulnerability assessment costs more than pen testing

55. According to the slides, which statement about mitigation and remediation is correct?

a) Mitigation removes the threat; remediation reduces impact
b) Both do exactly the same thing
c) Mitigation reduces impact; remediation removes the threat
d) Remediation never applies to software

56. CNA stands for:

a) Common Network Access
b) CVSS Notification Agent
c) Central Network Authority
d) CVE Numbering Authority

57. The National Vulnerability Database (NVD) is maintained by:

a) NIST
b) MITRE
c) The U.S. Department of Homeland Security
d) Microsoft

58. Which of these is a Temporal CVSS metric?

a) Attack vector
b) Remediation level
c) Scope
d) Confidentiality impact

59. Air-gapping a server (disconnecting it from external networks) lowers which metric?

a) Report confidence
b) Exploit code maturity
c) Attack Vector
d) Remediation level

60. Microsoft's 'Patch Tuesday' refers to:

a) The first Monday of each month
b) The last Friday of each month
c) Every day at midnight
d) The second Tuesday of each month

Answer Key: Set 6 (Questions 51 to 60)

Q	Answer	Why
51	**c) Strong, unique passwords for every account**	Weak and reused passwords are a cause; strong unique passwords are not.
52	**d) Miss most vulnerabilities inside the target environment**	Without login, they give an outsider's view and tend to miss internal issues.
53	**a) An exercise that tries to exploit vulnerabilities to measure real impact**	Pen tests exploit weaknesses to show what an attacker could achieve.
54	**b) Penetration testing exploits vulnerabilities**	Pen tests exploit; VA scans and prioritizes, and is cheaper.
55	**c) Mitigation reduces impact; remediation removes the threat**	Mitigation reduces impact; remediation completely removes the threat where possible.
56	**d) CVE Numbering Authority**	CNAs identify and distribute CVE IDs.
57	**a) NIST**	NVD is maintained by NIST and synchronized with the MITRE CVE list.
58	**b) Remediation level**	Temporal metrics: exploit code maturity, remediation level, report confidence.
59	**c) Attack Vector**	Air gapping removes remote exploitation, lowering the Attack Vector score.
60	**d) The second Tuesday of each month**	Microsoft usually releases patches on the second Tuesday.

SET 7 (Questions 61 to 70)

61. Which patch type adds new features and improves performance?

a) Feature update patch
b) Security patch
c) Bug fix patch
d) Rollback patch

62. A departing employee who takes sensitive materials is an example of which insider type?

a) Negligent worker
b) Departing employee
c) Security evader
d) Inside agent

63. Which is a recommended defence against insider attacks?

a) Disabling all logs
b) Giving administrator rights to everyone
c) Principle of least privilege
d) Sharing passwords within teams

64. A backdoor (or trapdoor) is:

a) Malware that encrypts files for ransom
b) A firewall rule for inbound traffic
c) A type of phishing email
d) A method that lets unauthorized users bypass security mechanisms

65. A key distinction of a logic bomb is that it:

a) Does not self-replicate
b) Always spreads over networks
c) Is a type of worm
d) Needs antivirus to activate

66. In the four phases of a virus, the phase that releases the malicious payload is:

a) Dormant phase
b) Execution (action) phase
c) Propagation phase
d) Triggering phase

67. A resident virus differs from a non-resident virus because it:

a) Infects files only when programs are not running
b) Changes browser homepages only
c) Embeds itself in RAM
d) Affects only the boot sector

68. A boot sector virus targets:

a) Browser settings
b) Email attachments
c) The file-sharing program
d) The master boot record

69. A polymorphic virus changes:

a) Its encryption key and decryptor, while its core function stays the same
b) Its entire code without any key
c) Only its file name
d) Nothing, since it is a worm

70. A metamorphic virus rewrites:

a) Only its decryption key
b) Its entire code, with no constant decryptor
c) Only the browser homepage
d) Only its host file name

Answer Key: Set 7 (Questions 61 to 70)

Q	Answer	Why
61	**a) Feature update patch**	Feature update patches add functions and improve efficiency.
62	**b) Departing employee**	Departing employees are a common insider threat type.
63	**c) Principle of least privilege**	Least privilege limits access to what each user needs.
64	**d) A method that lets unauthorized users bypass security mechanisms**	Backdoors bypass security to gain access.
65	**a) Does not self-replicate**	Logic bombs stay dormant until triggered and do not spread themselves.
66	**b) Execution (action) phase**	The execution phase releases the payload.
67	**c) Embeds itself in RAM**	Resident viruses stay in memory and keep infecting.
68	**d) The master boot record**	Boot sector viruses inject code into the master boot record.
69	**a) Its encryption key and decryptor, while its core function stays the same**	Polymorphic viruses change the encryption/decryptor layer per infection.
70	**b) Its entire code, with no constant decryptor**	Metamorphic viruses rewrite themselves without a fixed decryptor.

SET 8 (Questions 71 to 80)

71. A key limitation of signature-based antivirus is that it:

a) Cannot scan email attachments
b) Works only when offline
c) Cannot detect malware that has no existing signature
d) Detects all zero-day attacks

72. Unlike viruses, Trojan horses:

a) Always self-replicate
b) Attach only to boot sectors
c) Cannot steal data
d) Are usually stand-alone programs that do not inject into other programs

73. CryptoLocker, listed as a Trojan type in the slides, does what?

a) Encrypts files on the device and connected media
b) Gives remote administrative control
c) Steals banking details only
d) Floods a target with requests

74. What is the main difference between a virus and a worm?

a) A virus spreads over networks on its own
b) A worm spreads over networks on its own
c) A worm needs a host file to spread
d) A worm cannot replicate

75. Which statement correctly compares rootkits and Trojans?

a) A Trojan gives complete control; a rootkit gives limited access
b) Both give identical limited access
c) A rootkit gives complete control; a Trojan gives limited access
d) Rootkits are always visible to users

76. A dropper is malware that:

a) Removes antivirus software only
b) Acts as a botnet controller
c) Works as a firewall
d) Installs other malware onto a device

77. The main effect of bacteria or rabbit viruses is to:

a) Consume system resources until the system slows or crashes
b) Steal credit card numbers
c) Encrypt all files
d) Spread only by email

78. The Morris Worm, the first major Internet worm, appeared in:

a) 1974
b) 1988
c) 2000
d) 2017

79. The ILOVEYOU virus (2000) was:

a) A rootkit installed on ATMs
b) A Wi-Fi eavesdropping attack
c) Email-based, overwrote files, and spread globally
d) A DDoS attack on banks

80. The WannaCry worm (2017) exploited which Windows component?

a) SSH on Linux
b) An Oracle database service
c) The Apache web server
d) SMB (Server Message Block)

Answer Key: Set 8 (Questions 71 to 80)

Q	Answer	Why
71	**c) Cannot detect malware that has no existing signature**	Signature scanners only find known malware; zero-days slip through.
72	**d) Are usually stand-alone programs that do not inject into other programs**	Trojans are usually stand-alone and do not inject themselves.
73	**a) Encrypts files on the device and connected media**	CryptoLocker encrypts files and connected media.
74	**b) A worm spreads over networks on its own**	Worms are self-propagating; viruses need a host and user activity.
75	**c) A rootkit gives complete control; a Trojan gives limited access**	The slides contrast complete control (rootkit) with limited access (Trojan).
76	**d) Installs other malware onto a device**	Droppers are used in the first stage of an attack to install payloads.
77	**a) Consume system resources until the system slows or crashes**	They replicate without a payload, exhausting resources (DoS).
78	**b) 1988**	1974 Rabbit virus, 1988 Morris Worm, 2000 ILOVEYOU, 2017 WannaCry.
79	**c) Email-based, overwrote files, and spread globally**	ILOVEYOU spread through email and overwrote files.
80	**d) SMB (Server Message Block)**	WannaCry exploited Windows SMB.

SET 9 (Questions 81 to 90)

81. The controller of a botnet is called the:

a) Bot master (bot herder)
b) Search engine crawler
c) Chatbot assistant
d) Social media bot

82. Which of these is a beneficial bot?

a) A credential stuffing bot
b) Googlebot (web crawler)
c) A click fraud bot
d) An inventory hoarding bot

83. Which of these is malicious bot activity?

a) Indexing web pages
b) Answering routine customer questions
c) Credential stuffing
d) Posting weather updates

84. Spyware is best described as software that:

a) Shows unwanted advertisements
b) Encrypts files for ransom
c) Replicates across networks
d) Gathers private information and passes it to a third party

85. The purpose of adware is to:

a) Display unwanted advertisements to generate revenue
b) Steal passwords
c) Take control of the operating system
d) Replicate itself

86. Which of these is a countermeasure against malware?

a) Always running as administrator
b) Multi-factor authentication
c) Clicking pop-ups to close them
d) Disabling software updates

87. Using a non-administrator account helps because it:

a) Speeds up antivirus scans
b) Stops all email attacks
c) Limits the system changes a compromised user can make
d) Replaces the need for backups

88. According to the slides, the annual economic damage from malware worldwide is:

a) Under $1 million
b) Over $1 trillion
c) Exactly $10.5 trillion
d) Over $13 billion

89. According to the slides, what share of computer malware is made up of Trojans?

a) 58%
b) 28%
c) 72%
d) 85%

90. According to the slides, how many new pieces of malware are detected every day?

a) 56,000
b) 560,000
c) 5.6 million
d) 560 million

Answer Key: Set 9 (Questions 81 to 90)

Q	Answer	Why
81	**a) Bot master (bot herder)**	The bot master or bot herder controls the infected devices.
82	**b) Googlebot (web crawler)**	Web crawlers like Googlebot index web content.
83	**c) Credential stuffing**	Credential stuffing is an attack using automated login attempts.
84	**d) Gathers private information and passes it to a third party**	Spyware collects private data without the user's knowledge.
85	**a) Display unwanted advertisements to generate revenue**	Adware is advertising-supported software.
86	**b) Multi-factor authentication**	MFA adds an extra layer of account security.
87	**c) Limits the system changes a compromised user can make**	Admin accounts have full permissions that affect other users.
88	**d) Over $13 billion**	The slides give over $13 billion per year.
89	**a) 58%**	Trojans account for 58% of all computer malware (per the slides).
90	**b) 560,000**	The slides state 560,000 new malware detections per day.

SET 10 (Questions 91 to 100)

91. NIST defines cloud computing as:

a) A dedicated server for one organization
b) Software installed only on local PCs
c) On-demand network access to a shared pool of configurable computing resources
d) A hardware-only rental service

92. The 1965 business model by Corbato et al. proposed:

a) Free public web hosting
b) Local mainframe ownership
c) Mobile app development
d) Pay-as-you-go on-demand computing

93. Cloud computing emerged as a distributed model around:

a) 2005 to 2006
b) 1965
c) 1990
d) 2015

94. Which of these is NOT one of the five essential characteristics of cloud computing?

a) Measured service
b) Mandatory hardware purchase
c) Rapid elasticity
d) Resource pooling

95. Measured service means that:

a) Services are accessible on only one device
b) Resources are fixed monthly regardless of use
c) Usage is monitored and metered, such as by CPU hours
d) Customers manage all hardware

96. Dropbox is given in the slides as an example of which service model?

a) PaaS
b) IaaS
c) Edge computing
d) SaaS

97. Google App Engine is given as an example of which service model?

a) PaaS
b) SaaS
c) IaaS
d) Fog computing

98. In an IaaS model, the consumer controls:

a) Only the web browser interface
b) Operating systems and deployed applications
c) Nothing beyond the user account
d) The physical data center building

99. A hybrid cloud is best described as:

a) A public cloud used by one organization only
b) A cloud run by a single government
c) A mix of on-premises or private cloud and public cloud, with orchestration between them
d) A cloud with no internet connection

100. A community cloud is:

a) Used by one company only
b) Open to the general public on a pay-per-use basis
c) Not connected to any other cloud
d) Shared by a limited group of organizations with common goals

Answer Key: Set 10 (Questions 91 to 100)

Q	Answer	Why
91	**c) On-demand network access to a shared pool of configurable computing resources**	This is the NIST definition summarized in Class 4.
92	**d) Pay-as-you-go on-demand computing**	Corbato et al. proposed pay-as-you-go on-demand computing in 1965.
93	**a) 2005 to 2006**	The slides date the emergence of cloud computing to 2005-2006.
94	**b) Mandatory hardware purchase**	Cloud avoids hardware purchase; the five are self-service, network access, pooling, elasticity, and metering.
95	**c) Usage is monitored and metered, such as by CPU hours**	Metering tracks storage, CPU hours, and bandwidth to drive billing and scaling.
96	**d) SaaS**	SaaS gives applications over the web, such as Dropbox.
97	**a) PaaS**	PaaS provides a development environment, such as Google App Engine.
98	**b) Operating systems and deployed applications**	IaaS consumers manage the OS, storage, and applications.

Q	Answer	Why
99	**c) A mix of on-premises or private cloud and public cloud, with orchestration between them**	Hybrid combines private and public platforms.
100	**d) Shared by a limited group of organizations with common goals**	Community clouds serve a limited group, often for joint projects.

SET 11 (Questions 101 to 110)

101. A private cloud is:

a) Used exclusively by one organization
b) Shared with the public for a fee
c) Owned by a community of universities
d) Mixed with public services only

102. The cloud carrier role provides:

a) Physical servers in a data center
b) The network connectivity that carries cloud services
c) Legal ownership of consumer data
d) A web browser for end users

103. Which technology abstracts CPU, memory, and storage from applications and users?

a) Broadband access
b) Web technology
c) Virtualization
d) Storage area networks

104. Multi-tenant technology means:

a) One user runs many applications on separate hardware
b) Each tenant gets a private physical server
c) Users pay a fixed monthly fee for hardware
d) The same application logic serves many users at the same time

105. Which benefit does 'open standards' provide as a driver of cloud adoption?

a) A modular architecture that grows and changes as needed
b) Lower hardware prices by law
c) A guarantee of zero outages
d) No need for internet access

106. Which driver refers to avoiding capital expenditure on hardware?

a) Economies of scale
b) Small initial investment and low ongoing costs
c) Sustainability
d) Open standards

107. Edge computing is best described as:

a) Hosting all data in one central data center
b) A type of hardware failure
c) Hosting applications closer to end users or devices
d) A replacement for SaaS

108. In the slides, fog computing sits:

a) Inside the user's browser
b) Only in the central data center
c) Below the device layer as a sensor
d) Between the edge and the cloud

109. In the smart-camera example, the main benefit of edge processing is:

a) Less bandwidth, because footage is analyzed locally
b) More footage sent to the cloud
c) Cameras need no computing power
d) The cloud server must process all footage

110. Which statement is a characteristic of edge computing compared with cloud computing?

a) It needs constant internet for all processing
b) It can work without internet connectivity
c) It is centralized in far-away data centers
d) It is best for in-depth long-term analysis

Answer Key: Set 11 (Questions 101 to 110)

Q	Answer	Why
101	**a) Used exclusively by one organization**	A private cloud is used exclusively by one organization or its customers.
102	**b) The network connectivity that carries cloud services**	The carrier is the network link between provider and consumer.
103	**c) Virtualization**	Virtualization abstracts compute resources and enables multi-tenancy.
104	**d) The same application logic serves many users at the same time**	Multi-tenancy shares one application and infrastructure across users.
105	**a) A modular architecture that grows and changes as needed**	Open standards support modular, flexible architectures.
106	**b) Small initial investment and low ongoing costs**	Public cloud avoids buying hardware, software, or network devices.
107	**c) Hosting applications closer to end users or devices**	Edge moves computation closer to the data source.
108	**d) Between the edge and the cloud**	Fog is a layer between edge devices and the cloud.
109	**a) Less bandwidth, because footage is analyzed locally**	Motion detection at the edge means most footage never travels to the cloud.
110	**b) It can work without internet connectivity**	Edge can keep working offline; cloud needs connectivity.

SET 12 (Questions 111 to 120)

111. Vendor lock-in is:

a) Two providers sharing one server
b) Free migration between clouds
c) Dependence on one provider, making a switch costly or difficult
d) A backup policy

112. Portability, as defined in the slides, means:

a) The ability to share a server with others
b) Lowering internet costs
c) Access from mobile devices only
d) The ability of an application to move across environments

113. Which cloud challenge is linked to the lack of standard APIs and interfaces?

a) Interoperability
b) Performance and bandwidth cost
c) Recovery and loss of data
d) Reliability and availability

114. Which of these is one of the CSA 2026 eleven leading cloud threats?

a) Port scanning
b) Insecure APIs and interfaces
c) Data reduplication
d) Data scavenging

115. According to the report, which cloud threat ranks first?

a) Cloud misconfiguration
b) DDoS attacks
c) Inadequate Identity and Access Management
d) Insider threats

116. The slides rate cloud misconfiguration as a:

a) Low risk
b) Medium risk
c) No risk
d) High risk

117. The slides rate IAM attacks as a:

a) Critical risk
b) High risk
c) Medium risk
d) Low risk

118. The principle of least privilege means:

a) All staff get administrator rights
b) Users get only the permissions their role requires
c) All staff share one account
d) MFA is disabled for speed

119. The recommended protection for cloud misconfiguration is:

a) Endpoint Detection and Response (EDR)
b) Web Application Firewall (WAF)
c) Cloud Security Posture Management (CSPM)
d) Content Delivery Network (CDN)

120. Which protection is recommended for insecure APIs?

a) Immutable backups
b) Cryptojacking detectors
c) Hypervisor hardening
d) API gateway and WAF

Answer Key: Set 12 (Questions 111 to 120)

Q	Answer	Why
111	**c) Dependence on one provider, making a switch costly or difficult**	Lock-in makes switching providers expensive or technically hard.
112	**d) The ability of an application to move across environments**	Portability is a form of reusability across environments.
113	**a) Interoperability**	The industry lacks interoperability and API standards.

Q	Answer	Why
114	b) Insecure APIs and interfaces	Port scanning, data reduplication, and data scavenging are component attacks, not CSA threats.
115	c) Inadequate Identity and Access Management	The report states inadequate IAM ranks first.
116	d) High risk	Misconfiguration is rated high risk and is a common incident cause.
117	a) Critical risk	IAM attacks are rated critical risk.
118	b) Users get only the permissions their role requires	Least privilege limits each user to what they need.
119	c) Cloud Security Posture Management (CSPM)	CSPM identifies and fixes configuration weaknesses.
120	d) API gateway and WAF	An API gateway and WAF protect the API layer.

SET 13 (Questions 121 to 130)

121. A recommended protection against cloud ransomware is:

a) Immutable backups
b) Public storage buckets
c) Disabled audit logs
d) Shared administrator accounts

122. DNS amplification, a DDoS type, works by:

a) Stealing encryption keys
b) Using DNS infrastructure to amplify traffic
c) Reusing stolen tokens
d) Gaining physical access

123. An insider whose employee account is taken over by an attacker is a:

a) Malicious insider
b) Negligent insider
c) Compromised insider
d) Departing insider

124. VM escape means:

a) A VM moves to another data center
b) An administrator deletes a VM
c) A VM is backed up to tape
d) An attacker crosses the isolation boundary from a VM toward the hypervisor or host

125. Which is an example of a supply chain attack in the cloud?

a) A compromised CI/CD pipeline
b) A strong password policy
c) Regular backups
d) A physical lock on the server room

126. A typical APT activity is:

a) Immediate ransomware encryption
b) Persistence through cloud identities
c) A single short DDoS burst
d) Defacing a public website

127. SBOM and SCA are used to protect against:

a) DDoS attacks
b) Cryptojacking
c) Supply chain attacks
d) Insider threats

128. A consequence of cryptojacking is:

a) Lost email messages
b) Plain-text storage of files
c) Broken public websites only
d) Higher cloud bills and reduced processing capacity

129. Which mechanism protects data in transit according to the slides?

a) TLS 1.3
b) AES-256 encryption at rest
c) Tokenization
d) Database encryption

130. Tokenization works by:

a) Encrypting the whole network
b) Replacing sensitive values with tokens
c) Blocking DDoS traffic
d) Counting CPU hours

Answer Key: Set 13 (Questions 121 to 130)

Q	Answer	Why
121	**a) Immutable backups**	Immutable and offline backups make recovery possible.
122	**b) Using DNS infrastructure to amplify traffic**	It abuses DNS responses to multiply traffic.
123	**c) Compromised insider**	A compromised insider has an account hijacked by an outsider.
124	**d) An attacker crosses the isolation boundary from a VM toward the hypervisor or host**	VM escape breaks the isolation between VMs and the host.
125	**a) A compromised CI/CD pipeline**	CI/CD pipelines are a supply chain entry point.
126	**b) Persistence through cloud identities**	APTs keep long-term access, often through valid identities.
127	**c) Supply chain attacks**	They help track and check third-party software components.
128	**d) Higher cloud bills and reduced processing capacity**	Cryptojacking consumes resources, raising bills and slowing services.
129	**a) TLS 1.3**	TLS protects data moving across a network.
130	**b) Replacing sensitive values with tokens**	Tokens stand in for sensitive values such as card numbers.

SET 14 (Questions 131 to 140)

131. Port scanning belongs to which cloud component category?

a) Storage (A3)
b) Applications (A4)
c) Network (A1)
d) Virtual machines (A2)

132. A cross-VM side channel attack is an example of an attack on:

a) Storage (A3)
b) Network (A1)
c) Applications (A4)
d) Virtual machines (A2)

133. Data scavenging is an attack on:

a) Storage (A3)
b) Network (A1)
c) Virtual machines (A2)
d) Applications (A4)

134. Malware injection is an attack on:

a) Network (A1)
b) Applications (A4)
c) Storage (A3)
d) Virtual machines (A2)

135. The Capital One breach in 2019 affected approximately:

a) 8.5 million devices
b) 91.7 GB of data
c) 106 million individuals
d) 2,700 organizations

136. The vulnerability Capital One's attacker exploited was:

a) A zero-day in CrowdStrike
b) A stolen signing key
c) Password spraying
d) Server-side request forgery (SSRF)

137. The Storm-0558 campaign relied on:

a) Theft of a signing key and forged authentication tokens
b) SMS phishing against MFA
c) SQL injection of databases
d) Malicious VM images

138. Midnight Blizzard's initial access to Microsoft used:

a) Physical access to data centers
b) Password spraying against a legacy test tenant
c) A malicious USB drop
d) A faulty software update

139. The Snowflake breaches of 2024 were linked to:

a) AWS internal systems only
b) A hardware recall
c) Stolen credentials used on accounts without MFA
d) Physical theft of servers

140. The European Commission breach of 2026 was caused by:

a) Stolen laptops
b) A hypervisor escape
c) A DNS hijack of a public site
d) A Trivy software supply chain compromise

Answer Key: Set 14 (Questions 131 to 140)

Q	Answer	Why
131	**c) Network (A1)**	Port scanning is a network-based attack.

Q	Answer	Why
132	d) Virtual machines (A2)	VM attacks include cross-VM side channels.
133	a) Storage (A3)	Data scavenging searches storage residue for sensitive data.
134	b) Applications (A4)	Malware injection inserts code into cloud applications.
135	c) 106 million individuals	About 106 million people in the USA and Canada.
136	d) Server-side request forgery (SSRF)	SSRF led to AWS instance metadata and temporary credentials.
137	a) Theft of a signing key and forged authentication tokens	Attackers forged tokens after stealing a Microsoft signing key.
138	b) Password spraying against a legacy test tenant	Password spraying hit a legacy non-production tenant.
139	c) Stolen credentials used on accounts without MFA	Some compromised accounts did not have MFA enabled.
140	d) A Trivy software supply chain compromise	A compromised Trivy scanning tool exposed an AWS secret.

SET 15 (Questions 141 to 150)

141. The CrowdStrike outage of July 2024 was:

a) A defective software update, not a cyberattack
b) A nation-state attack
c) A ransomware encryption event
d) A stolen AWS key

142. According to the slides, about how many Windows devices did the CrowdStrike outage affect?

a) 106 million
b) 8.5 million
c) 91.7 GB
d) 2,700

143. The Azure disruption of 18 July 2024 lasted approximately:

a) Five years
b) Under one minute
c) Half a day
d) A full week

144. The Toyota cloud data exposure of 2023 was caused by:

a) A zero-day in a hypervisor
b) Phishing of an administrator
c) A botnet attack
d) Cloud configuration errors that made data publicly accessible

145. The Football Australia AWS incident of 2024 involved:

a) Exposed AWS credentials
b) Stolen laptops
c) A hypervisor escape
d) A DDoS against a website

146. ChaosDB (2021) was a vulnerability in:

a) AWS S3
b) Azure Cosmos DB that could expose database credentials
c) Google Drive
d) Snowflake SQL queries

147. The Code Spaces attack of 2014 involved:

a) Customer data sold on the dark web
b) Ransomware on Windows PCs
c) An attacker taking control of the AWS console and deleting resources
d) A DNS amplification attack

148. The Retool compromise of 2023 succeeded because attackers:

a) Found a public S3 bucket
b) Injected code into a CI/CD pipeline
c) Exploited a hypervisor bug
d) Used SMS phishing and bypassed MFA

149. Which of these is one of the five key lessons from the incidents?

a) Insufficient monitoring: watch API calls and access-key creation
b) Monitoring is unnecessary in the cloud
c) Backups are needed only on premises
d) Supply chain risk is minimal

150. The Cisco CEO quoted in the slides said cloud computing:

a) Removes all security risks
b) Is a security nightmare that cannot be handled in traditional ways
c) Is safer than all on-premises systems
d) Needs no service level agreement

Answer Key: Set 15 (Questions 141 to 150)

Q	Answer	Why
141	**a) A defective software update, not a cyberattack**	It was a defective security update with worldwide impact.
142	**b) 8.5 million**	The slides say about 8.5 million devices.
143	**c) Half a day**	The report says the disruption lasted about half a day.
144	**d) Cloud configuration errors that made data publicly accessible**	The incident was a misconfiguration.
145	**a) Exposed AWS credentials**	Exposed credentials raised concerns about storage access.
146	**b) Azure Cosmos DB that could expose database credentials**	A Cosmos DB notebook feature flaw; Microsoft remediated it.
147	**c) An attacker taking control of the AWS console and deleting resources**	The attacker deleted resources, and the company ceased operations.
148	**d) Used SMS phishing and bypassed MFA**	Attackers phished employees and bypassed MFA weaknesses.
149	**a) Insufficient monitoring: watch API calls and access-key creation**	The report stresses monitoring API calls, admin activity, and key creation.
150	**b) Is a security nightmare that cannot be handled in traditional ways**	The slides paraphrase John Chambers' view.

Study tip: the question order mixes topics on purpose. Re-do any set where you scored below 8/10, and revisit the matching class notes. Figures such as scores, counts, and dates come from the course slides; verify them against the original sources before the exam.